



Armenian Epigraphy and Cultural Heritage Data Analysis

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Abstract

Building searchable corpora of Armenian epigraphy is challenging due to OCR noise, various book layouts, and Armenian’s rich morphology. This paper proposes a reproducible workflow for converting scanned epigraphic volumes into structured inscription records and for detecting inscription-type terminology in running text. This work combine layout-aware pre-processing and inscription extraction with supervised NER trained from an expert term list. On held-out books, transformer models generalize well to previously seen term types in new grammatical contexts, but do not reliably identify truly unseen terms, motivating future work in context-heavy discovery and expert-assisted lexicon expansion.

1 Introduction

1.1 Motivation

Armenian epigraphy constitutes one of the most extensive and historically significant inscriptional traditions of the Caucasus, spanning many centuries and a broad geographic distribution. These inscriptions serve not only as primary historical sources but also as critical witnesses to Armenian presence beyond Armenian borders. Despite their scholarly importance, a large portion of the Armenian epigraphic corpus remains accessible primarily through printed editions and scanned books, which limits their usability for large-scale computational analysis and automatic knowledge extraction.

1.2 Challenges

The digitization of Armenian epigraphic material introduces a series of technical challenges that go beyond simple optical character recognition (OCR). The source data are characterized by heterogeneous layouts, complex typographical conventions, frequent OCR errors, and the rich morphological structure of the Armenian language. These factors make direct application of standard natural language processing (NLP) tools unreliable without domain-specific adaptation. At the same time, recent advances in named entity recognition (NER) and transformer-based language models offer new possibilities for extracting structured information from large historical text collections, even in low-resource language settings.

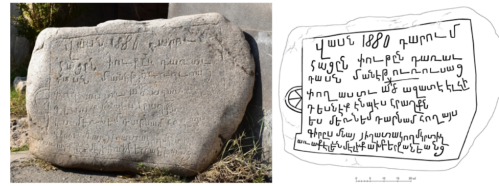


Figure 1: Rock inscription of Vahrvavar (Figure 5a, b in Հարությունյան, Արսեն [2023a]).

1.3 Goals

This report presents a practical pipeline for transforming digitized Armenian epigraphic sources into a machine-interpretable corpus and for applying NER techniques to automatically detect and most importantly *discover* inscription-related terminology. The work focuses on three main stages: (1) the pre-processing and cleaning of OCR-produced Armenian text, (2) the structured extraction of inscription units and their associated metadata, and (3) the construction and evaluation of NER models for identifying epigraphic terms in running text. The extracted entities are subsequently used as the basis for building a concise, data-driven lexicon of Armenian epigraphic terminology.

By reporting on the full workflow; from data acquisition to model evaluation on unseen corpora, this work aims to provide a reproducible reference for future computational studies of Armenian epigraphy and, more broadly, for low-resource historical corpora with complex internal structure.

2 Related Work

One of the most influential frameworks for the standardized description of inscriptions in digital epigraphy is the **EAGLE model** (Electronic Archive of Greek and Latin Epigraphy). Developed within a series of European research initiatives, EAGLE provides a shared conceptual and technical infrastructure for the aggregation, description, and interoperability of epigraphic data across institutions. At its core, the EAGLE model proposes a structured vocabulary and metadata schema that enables consistent documentation of inscriptions, regardless of their geographic origin, material support, or historical period.

This archive is taken as a reference for this study, as the ultimate goal would be to obtain a well-defined and hierarchically-coherent lexicon built around seven key categories *Material*, *Execution Technique*, *Type of Inscription*, *Object*

Type, Decoration, Dating Criteria, and State of Preservation. This categorization would allow the implementation of Armenian epigraphy into more general infrastructures.

3 Data

The dataset is composed of two complementary building blocks: an OCR-derived corpus of inscription texts and an expert-curated list of epigraphic terminology.

3.1 Inscription Corpus

The corpus treated throughout this study is a compilation of methodically chosen books that each document an array of inscriptions in full, representing an accurate and dense data resource. This type of book was chosen given that each groups a considerable quantity of information in a concise format, optimizing the pre-processing workflow by eliminating the need for spatial aggregation.

The book selection was tailored to meet four criteria that ensure both cultural and conceptual completeness, namely authorship diversity, genre balance, regional and finally temporal coverage.

Table 1 extensively presents the different members of the corpus. The main series of works is *Divan Hay Vimagrutyun* (DHV) (Barkhudaryan [1967, 1973, 1982], Barkhudaryan et al. [2012], Grigoryan [1996], Orbeli [1965, 1960], Sargsyan et al. [2017], Avagyan and Janpoladyan [1977], Grigoryan [1999]), given that it is one of the most detailed and overall consistent corpus Armenian Inscriptions. In addition to the DHV works, multiple other regional corpora lead by specialists such as Karapetyan, Kortoshyan, Petrosyan, Barkhudaryan, and Harutyunyan were considered to encapsulate the diversity of Armenian Epigraphy (Karapetyan [2021a,b], Քորթոշյան, Րաֆֆի [2024], Kortoshian [2013], Petrosyan [2008], Barkhudaryan [1999], Harutyunyan [2022], Harutyunyan).

The complete set of volumes was then digitized and converted into machine-readable text using an Armenian-adapted OCR pipeline, forming the textual basis for all subsequent pre-processing and annotation steps.

3.2 Inscription Terminology Expert List

The scope of this study is focused on the *Types of Inscriptions* domain, representing the

function of inscriptions, closely linked to the material, linguistic and historical background of the inscription. The development of this list was done as follows: a team of specialists first compile a preliminary collection of terms, which was itself augmented using generative AI methods (GPT-5). Then, the augmented part was then manually reviewed by the experts who shortlisted a total of **41 terms**. These include general descriptors such as *վիմագիր* and *վիմակալն արձանագրություն*; functional classes such as funerary (*տապանագիր*), donor (*նվիրագիր*), commemorative (*հուշագիր*), legal (*կանոնագիր*), and construction-related inscriptions (*շինարարական արձանագրություն*) amongst other characteristics.

3.3 pre-processing

Once the books have been digitized, the next step is to design a robust pre-processing pipeline capable of segmenting the input texts into individual sentences. The primary challenges arise from the structure of the raw OCR output. First, the OCR tends to fragment paragraphs into shorter, often incomplete units separated by unintended line breaks. Second, it frequently misrecognizes the Armenian full-stop character (.), replacing it with a colon (:) or other punctuation. These issues complicate both the detection and preservation of the texts' logical and syntactic integrity. Consequently, a careful understanding of the organizational conventions of the source material is essential for constructing an effective pre-processing workflow.

3.3.1 Paragraph Structuring and Parsing

The corpus consists of books that compile multiple epigraphic works. Each book follows a consistent internal structure that can be leveraged during pre-processing. Pages are delineated by explicit markers. Within each page, inscriptions are numbered and labeled by artefact, followed by a brief description indicating the location of the inscription and the number of lines it spans. This line count is typically identifiable through the Armenian word for "line" (*տող*) or one of its synonyms. The inscription itself follows, written in uppercase letters. After the inscription, a descriptive paragraph commonly appears, often beginning with the abbreviation "ծանոթ." ("familiarization"), as well as a short note referencing the original publication from which the inscription was sourced.

Book	Abbrev.	Citation
Divan Hay Vimagrut' yan 1	DHV-1	Orbeli [1965]
Divan Hay Vimagrut' yan 2	DHV-2	Orbeli [1960]
Divan Hay Vimagrut' yan 3	DHV-3	Barkhudaryan [1967]
Divan Hay Vimagrut' yan 4	DHV-4	Barkhudaryan [1973]
Divan Hay Vimagrut' yan 5	DHV-5	Barkhudaryan [1982]
Divan Hay Vimagrut' yan 6	DHV-6	Avagyan and Janpoladyan [1977]
Divan Hay Vimagrut' yan 7	DHV-7	Grigoryan [1996]
Divan Hay Vimagrut' yan 8	DHV-8	Grigoryan [1999]
Divan Hay Vimagrut' yan 9	DHV-9	Barkhudaryan et al. [2012]
Divan Hay Vimagrut' yan 10	DHV-10	Sargsyan et al. [2017]
Historical Monuments of Akunk and Katnaghybur	KAR-2014	Կարապետյան, Սամվել [2014]
Khodjavank Monastery	KAR-2024a	Կարապետյան, Սամվել [2024]
Northern Artsakh	KAR-2021a	Karapetyan [2021b]
Lapidary Inscriptions of Bun Aghvank	KAR-2021b	Karapetyan [2021a]
Armenian Collection of the Caucasian Museum	KAR-2004	Կարապետյան, Սամվել [2004]
Previously Unpublished Inscriptions of Yerevan Katoghike	HAR-2019a	Հարությունյան, Արսեն [2019a]
Epigraphic Heritage of Armavir Province	HAR-2017	Հարությունյան, Արսեն [2017]
Newly Discovered Tombstones of Holy Ejmiatsin	HAR-2021	Հարությունյան, Արսեն; Մելիքյան, Արմինե [2021]
Settlement of Noragavit and St. Gevorg Church	HAR-2019b	Հարությունյան, Արսեն [2019b]
Inscribed Artefacts of the N. Adonts Museum of Sisian	HAR-2019c	Հարությունյան, Արսեն [2019c]
Epigraphic Heritage of Tatev Hermitage	HAR-2023	Հարությունյան, Արսեն [2023b]
Natural Disasters and Celestial Phenomena in Inscriptions	HAR-2022/2023	Հարությունյան, Արսեն [2022, 2023a]
Melik Mansions of Artsakh and Syunik	GHU-2001	Դուլյան, Արտակ [2001]
Inscriptions of Armenian Settlements in India	KOR-2024	Թորոնշյան, Բաֆֆի [2024]
Inscriptions of Aleppo	KOR-2013	Kortoshian [2013]
Tsaghatskar Monastery: Historical and Archaeological Study	MEL-2024	Մելիքյան, Հուսիկ [2024]
Birthplaces of Genocide Survivors in Lebanese Funerary Inscriptions	TAS-2018	Tashjian [2018]
Georgian State Policy and Armenian Cultural Monuments (1988–1998)	KAR-1998	Karapetyan [1998]

Table 1: Publications included in the corpus, with abbreviations used throughout the paper. Light-gray rows indicate the volumes used to construct the full training dataset for the main experiments. The earliest baseline model was trained solely on **DHV-10**. The remaining publications (non-highlighted rows) were held out as unseen material for evaluating the model’s ability to generalize and to detect potential new inscription-type terminology.

The paragraph-restructuring procedure therefore proceeds in several stages:

1. segment the text by page,
2. further divide each page into paragraphs while preserving the underlying structural markers,
3. repair fragmented sentences and correct intra-paragraph OCR inconsistencies, and
4. reconstruct fully coherent pages to produce a machine-interpretable corpus.

3.3.2 Data Cleaning

Prior to executing the inscription extraction and named entity recognition pipelines, a dedicated data cleaning stage was implemented to remove structurally irrelevant pages, mitigate OCR-induced errors, and stabilize the statistical properties of the corpus. A superficial cleaning strategy proved insufficient, as the digitized books contain a substantial amount of erroneously extracted metadata, particularly at their beginnings and ends. Consequently, a two-stage cleaning framework was devised, consisting of an initial sentence-level filtering pass followed by a finer-grained token-level refinement. This design enables effective noise suppression while

ensuring that the cleaning procedure remains scalable and independent of corpus-specific formatting.

As shown in Figure 3, the majority of sentences in the corpus contain fewer than five words, with a large proportion of these being genuinely empty or non-informative (Figure 2). To retain as much informative content as possible while eliminating spurious entries, all sentences shorter than three words were discarded, resulting in the removal of approximately 27% of the raw data.

In the second, more granular cleaning stage, the character length of the longest token within each sentence was computed. Figure 4, which depicts the distribution of these maximum token lengths, reveals that only a small fraction of sentences exhibit a longest token shorter than five characters. This threshold was not selected arbitrarily: it corresponds to the length of the shortest reference epigraphic term and was therefore adopted as the lower bound for token-level filtering, allowing the removal of linguistically implausible content while preserving all potentially valid inscriptions. This cleaning step reduces the number of extracted sentences from 157,329 to 99,330 samples.

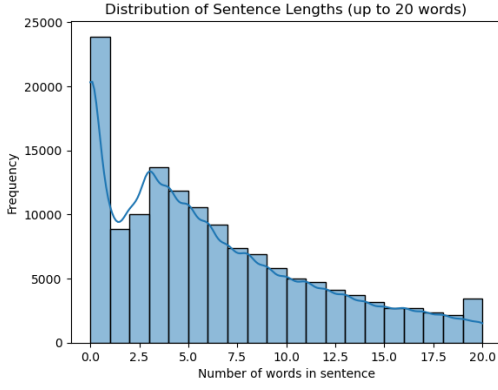


Figure 2: Distribution of sentence lengths under 20 words.

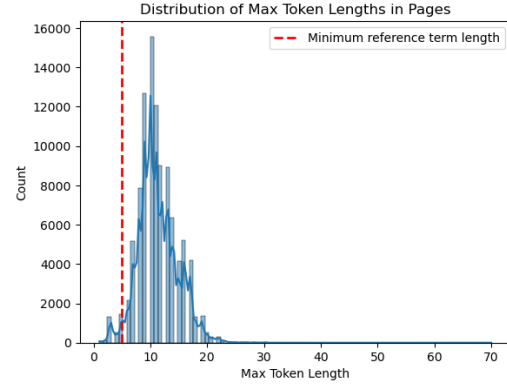


Figure 4: Distribution of Longest Tokens' Lengths in Pages.

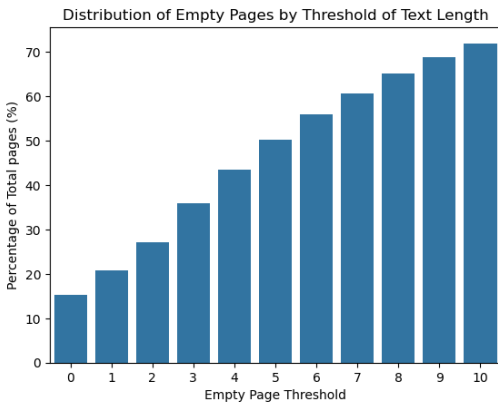


Figure 3: Cumulative sum of short sentences by threshold of maximum length.

3.4 Inscription Extraction

Following the restructuring of paragraphs and the restoration of syntactic continuity, the next objective is to systematically extract the inscriptions and their associated metadata into a structured, machine-readable format. Ultimately, the goal is to produce a dedicated dataframe for each corpus, where every row corresponds to a single inscription enriched with the relevant contextual information, such as its numbering, the artefact label, location description, line count, and accompanying scholarly notes.

To achieve this, we rely on the characteristic internal organization of the texts. As discussed earlier, inscriptions typically appear after a descriptive header indicating the artefact, followed by keywords related to location or line span (commonly containing terms such as տող “line”). They are written in uppercase and usually follow by commentary beginning with markers such as Ծանոթ. or references to original publications (e.g., Հրատ.).

We developed two complementary extraction strategies, as described below.

Method 1: Keyword-based Detection This approach exploits the semi-standardized internal structure that governs the majority of inscriptions in the corpus. In a first deterministic pass, the method identifies valid inscription candidates by detecting structured title lines through regular expressions, followed by the extraction of consecutive uppercase text blocks corresponding to the inscription body, and the attachment of associated publication and commentary lines marked by Հրատ. and Ծանոթ.. Page-level filtering based on description markers such as տող., Միատող., and տողից. is applied to restrict processing to relevant regions and reduce noise. To address OCR fragmentation, layout irregularities, and cross-page inscriptions, a second, fallback term-driven pass is executed, which ranks and tracks structural markers across pages, resolves consecutive inscription sequences, and merges split description–publication–note patterns when necessary. The combined two-pass strategy yields fully structured inscription records including page number, title, inscription text, publication reference, and scholarly notes. While this hybrid pipeline achieves high precision and robustness against OCR inconsistencies, its reliance on a predefined epigraphic layout makes it sensitive to inscriptions that deviate substantially from the expected structural format.

Method 2: Title-based Detection In contrast, this strategy focuses on identifying the titles or numbering conventions that precede inscriptions, for example, artefact numbers or section headers standardized across the corpus. Because

Venn Diagram of Divan Prak 10 Inscription Extractions

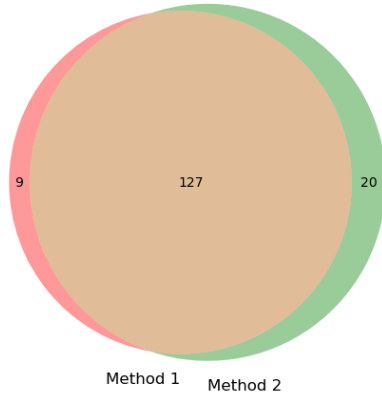


Figure 5: Venn Diagram of Total Extractions

Venn Diagram of Divan Prak 10 Inscription that have no MetaData

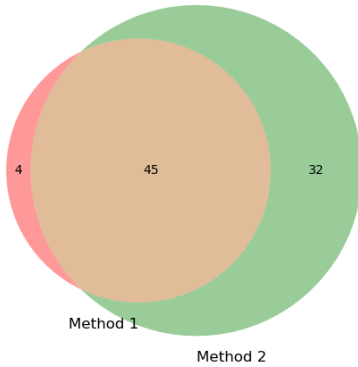


Figure 6: Venn Diagram of Complete Extractions

these titles appear with high regularity and clear visual structure, title-based detection scales effectively across large volumes of material and can recover a greater number of inscription candidates. The trade-off is that the extracted segments may be leaner, capturing the inscriptions themselves but occasionally omitting surrounding descriptive information.

In practice, these two approaches are complementary. Keyword-based detection excels at maximizing informational completeness, while title-based detection offers greater generalizability across diverse corpora. Depending on the downstream applications whether prioritizing completeness or coverage, it is advantageous to combine both methods within a hybrid extraction framework. This yields a robust set of inscription units, each paired with its structural metadata, which forms the basis of the final machine-interpretable dataset.

Figure 5 reports the total number of inscriptions detected by each method and their overlap. Out of 156 unique inscription candidates, 127 are detected by both methods, corresponding to an overlap of approximately 81.4%. Method 1 uniquely identifies 9 inscriptions (5.8%), while Method 2 uniquely recovers 20 inscriptions (12.8%). This indicates that although the two methods largely agree, Method 2 provides a clear gain in overall coverage, capturing more additional candidates that would otherwise be missed by keyword-based detection alone.

Figure 6 restricts the analysis to *complete* extractions, where both the inscription text and all associated metadata are successfully recovered. In this stricter setting, a total of 81 complete inscriptions are observed, of which 45 (55.6%) are shared by both methods. Method 1 contributes only 4 unique complete extractions (4.9%), whereas Method 2 contributes 32 unique complete extractions (39.5%). These results demonstrate that while Method 1 is effective for preserving contextual richness in overlapping cases, Method 2 is substantially more effective at scaling complete extractions across the corpus.

Overall, the numerical evidence confirms a clear division of strengths between the two strategies. Method 2 maximizes scalability and total yield, while Method 1 plays a complementary role by reinforcing metadata completeness in shared detections. Their combination therefore enables a hybrid extraction framework that simultaneously optimizes coverage and structured data quality, which is critical for constructing reliable machine-interpretable epigraphic corpora.

3.5 Named Entity Recognition and Lexicon Formation

Once the corpus has been segmented into paragraphs, a sentence splitter is designed and applied to the text. Standard sentence splitters such as the NLTK tokenizer do not perform reliably on Armenian linguistic data, so it is necessary to develop a custom method that relies on splitting sentences solely on the Armenian full stop mark $\therefore$. This step was conducted post-cleaning to ensure that all the text was denoised of OCR-related mistakes.

After compiling the corpus into individual sentences, the next objective is to build a model capable of detecting both known and previously unseen inscription-related terms. The chosen approach is to train an encoder-only, sequence-to-sequence transformer on a Named Entity Recog-

nition task. These models capture full contextual information by processing the input sequence of tokens in both directions without masking any part of the input. This property makes them well suited for the requirements of the task.

Named Entity Recognition is a standard NLP task that categorizes spans of text into meaningful classes. In this application, it is used to tag segments of each sentence with the appropriate inscription type.

3.5.1 Ground Truth Label Extraction

Training a model to identify terms of interest within a sentence requires a dataset of sentences that have been pre-tagged to reflect the expected ground-truth extractions. Given the corpus of sentences and a list of epigraphy-related reference terms, it is essential to construct a pipeline that can detect the occurrence of these terms and extract the associated metadata, including context, page number, and the title of the source book. To produce this ground-truth dataset, the extraction process relied on OpenAI's GPT-5 model. A predefined text chunk was supplied to the model together with the reference term list, using a prompt designed according to *chain-of-thought* principles¹. The prompt guides term extraction by focusing on predefined terms, recording each occurrence separately with surrounding context for clarity. Metadata like file name and page number ensure traceability. Extracted data is organized hierarchically, with occurrences grouped under terms and terms compiled into an extraction result. The prompt emphasizes Armenian language handling to account for inflectional and orthographic variations, without relying on explicit morphological rules, ensuring robust zero-shot extraction.

Finally, the most crucial part of this pipeline is the output formatting, which relied on Pydantic, a type-forcing python library, which is used in our case to pass a JSON structure to the prompt which directs the model to output coherent and reproducible results by forcing the output's structure.

Figure 7 illustrates an example of the structured JSON schema enforced through Pydantic and returned by the model. The root object contains a list of extracted terms, where each entry corresponds to a reference inscription term provided to the model. For each term, both its

```

1 {
2   "terms": [
3     {
4       "reference_term": "Շինարարական
5       արձանագրություն",
6       "definition": "Շինարարության կամ
7       վերանորոգման արձանագրություններ",
8       "extractions": [
9         {
10          "file_name": "Divan_Prak5",
11          "page_number": 118,
12          "term": "Շինարարական արձանագրություն",
13          "context": "Երրորդ շինարարական
14          արձանագրությունը պարզապես նորոգումներին է
15          վերաբերում"
16        },
17        {
18          "file_name": "Divan_Prak5",
19          "page_number": 123,
20          "term": "Շինարարական արձանագրություն",
21          "context": "Վանքի շինարարական
22          արձանագրության մեջ արդեն հիմնադիր
23          հովհաննեսը"
24        }
25      ]
26    }
27  ]
28 }

```

Figure 7: Example of structured JSON output for inscription term extraction.

canonical form and its definition are stored. The extractions field then aggregates all detected occurrences of the term within the processed corpus. Each extraction instance includes the source file name, the page number, the matched term, and the surrounding textual context in which the term appears. This hierarchical structure enables a one-to-many relationship between reference terms and their occurrences in the corpus, thereby ensuring traceability of each extraction back to its original document location. Furthermore, the strict typing enforced by the schema guarantees consistency across all generated samples, making the resulting dataset directly suitable for supervised learning and downstream evaluation tasks.

3.5.2 NER-fit Dataset Formation

Once the ground-truth extractions have been generated, the annotated data is converted into the standard Inside–Outside–Beginning (IOB) tagging scheme. This step introduces an additional challenge due to the grammatical flexibility of the Armenian language, where inflection and variable word order can obscure the boundaries of multiword terms and complicate token-level labeling.

After applying IOB tagging, the dataset is partitioned with the explicit goal of isolating the test labels from both the training and evaluation sets. This separation is essential for obtaining robust and unbiased results. While the training and

¹Specifically, we applied GPT-5 over predefined text chunks together with the reference term list. We used chain-of-thought prompting to improve reasoning as presented in Appendix A.

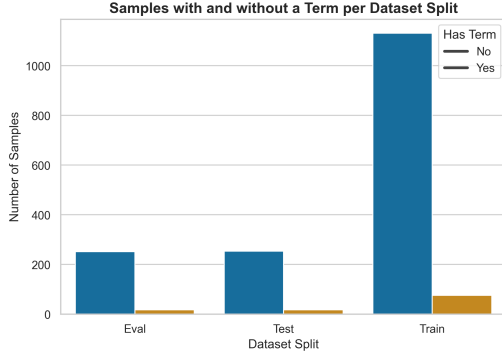


Figure 8: Baseline split distribution.

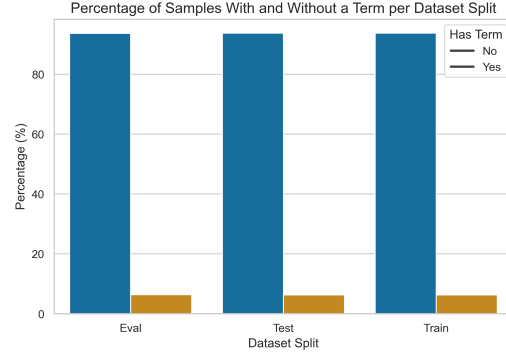


Figure 9: Baseline percentages of sentences that contain inscription-related terms and sentences that do not.

evaluation subsets may share certain term types, the test set consists exclusively of novel, previously unseen inscription-related terms, ensuring that the final performance metrics genuinely reflect the model’s ability to generalize beyond the vocabulary encountered during training.

Figures 8, 9 and 10 illustrate the structure of the splits. The distributions show that sentences containing terms are well represented and balanced across all subsets, maintaining a consistent 85/15 ratio within each split.

Figure 10 illustrates the terms’ frequency distribution across the three splits, illustrating the aforementioned overlap between the training and evaluation splits, and the relative isolation of the test split. It is also important to mention that the testing split is hermetic to both the grammatical and syntactical aspects of the training set, meaning that the terms in the testing set are neither synonyms nor grammatical adaptations of the training set’s.

This distribution is quantified in Table 3, approximately 5% of the terms used in training are seen at evaluation whereas none are used in testing, effectively leaving out the testing set. The same property holds for the restrained, baseline set (Table 2) where a larger 26% of training words are seen in evaluation.

3.5.3 Baseline Definition

A considerable challenge of this research quest is dealing with limited resources when it comes to machine learning models that are adequate for treating Armenian data. Paring the novelty of this task with the aforementioned constraint, it is thus important to establish preliminary results to assess the feasibility of this research.

The baseline experiment is designed to use a restrained subset of the total data to train and evaluate a lightweight, malleable model:

SpaCy, an industrial-strength NLP library designed for efficient training and inference, providing out-of-the-box pipelines for named entity recognition. Its NER component is particularly well suited for rapid experimentation, as it supports incremental training and straightforward integration of custom entity labels.

For this work, a spaCy pipeline was initialized and extended with a custom NER component capable of predicting inscription-related entity types. The training, evaluation and test set were formed using the DHV-10 samples (Table 1). Training followed a standard stochastic optimization procedure with shuffled batches and periodic evaluation. Early stopping was applied through a validation loss monitor to prevent overfitting, halting training once improvements plateaued.

After training, the model was evaluated on the held-out testing set using. Performance metrics included precision, recall, and F1-score for entity spans. This baseline provided a valuable reference point for assessing the improvements introduced by the transformer-based approach described in later sections.

Table 4 shows that while this model achieved a relatively high precision of 50%, indicating that the entities it predicted were often correct, its recall remained extremely low at around 6%, showing that it failed to identify the vast majority of true epigraphic terms; together, this imbalance resulted in a weak overall F1 score and revealed the model’s limitations in handling the sparse, morphologically complex terminology characteristic of Armenian inscriptions, ultimately underscoring the need for more context-sensitive architectures.

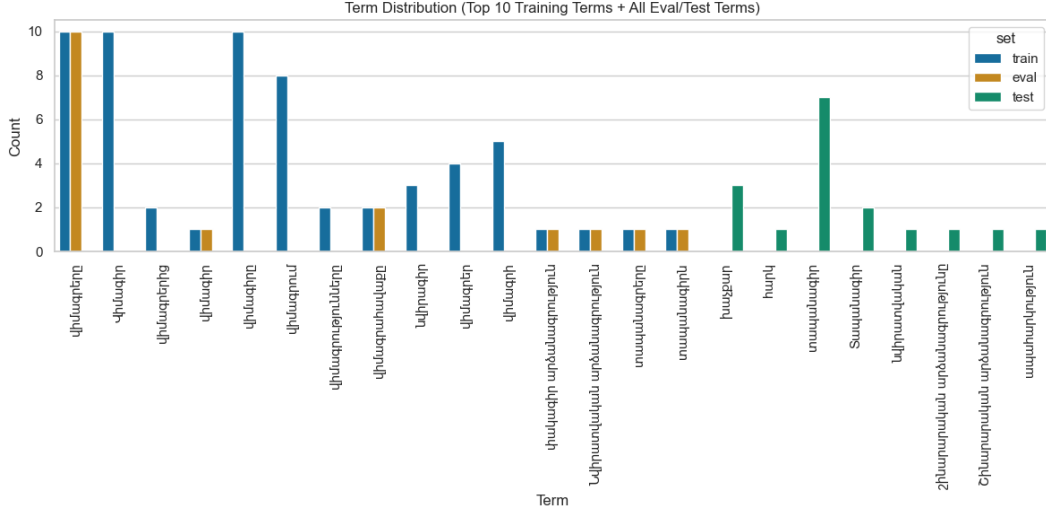


Figure 10: Distribution of terms’ frequencies across splits for baseline data (Top 10 Training Terms and all Evaluation and Test Terms).

Split	# Sentences	# Sentences w/ Term	# Unique Terms	# Overlap w/ Train
Train	1208	76	27	-
Development	269	17	7	7
Test	271	17	8	0

Table 2: Dataset statistics for the NER training, development, and test splits, drawn from **DHV-10**. The test set contains only unseen terms.

3.5.4 Expanding the Model Range

While spaCy provides a fast and accessible baseline for named entity recognition, its architecture and training resources limit its ability to capture the linguistic complexity of Armenian epigraphic texts. Armenian exhibits rich inflection, flexible word order, and domain-specific terminology that often appears in sparsely distributed or highly context-dependent forms. These characteristics make transformer-based models far better suited to the task. To address these challenges, three pre-trained transformer models were introduced into the evaluation pipeline.

The first model, **Multilingual BERT (mBERT)**, is trained on over one hundred languages, including Armenian. Its broad multilingual coverage makes it a strong general-purpose baseline for low-resource settings, as it benefits from shared subword representations across related and unrelated languages. This cross-lingual transfer often helps compensate for limited training data in resource-scarce domains such as epigraphy.

The second model, **HyeBERT**, is an Armenian-specific transformer pretrained exclusively on Armenian corpora. Unlike multilingual models, HyeBERT directly encodes

the morphology, syntax, and lexical structures unique to Armenian. This specialization makes it particularly well suited for tasks that require high-sensitivity to grammatical adaptation and context-encapsulation.

Finally, **XLM-RoBERTa**, a large multilingual model trained on massive CommonCrawl data, offers a more robust and modern alternative to mBERT. Its training procedure and scale provide stronger cross-lingual representations and improved contextual modeling. As a result, XLM-RoBERTa is well positioned to handle the variability and ambiguity present in Armenian text, even when specific epigraphic terms are rare or morphologically diverse.

The main technical characteristics of the transformer models retained for evaluation are summarized in Table 5. Although mBERT is trained on a slightly larger number of languages, XLM-RoBERTa differs in several key architectural and representational aspects, most notably in its tokenization strategy and parameterization. Both models adopt a 12-layer encoder with a hidden size of 768, but XLM-RoBERTa contains more than twice the number of parameters of mBERT (approximately 270M versus 110M).

Split	# Sentences	# Sentences w/ Term	# Unique Terms	# Overlap w/ Train
Train	51223	530	136	-
Development	269	17	7	7
Test	271	17	8	0

Table 3: Dataset statistics for the NER training, development, and test splits, drawn from **complete training set**. The test set contains only unseen terms.

Model Name	Precision	Recall	F1
Bert-Multilingual Uncased	0.78	0.39	0.52
XLNet-Roberta	0.78	0.39	0.52
Spacy	0.50	0.06	0.11
HyeBert	0.34	0.07	0.09

Table 4: Baseline model performance.

Moreover, mBERT relies on a WordPiece tokenizer, whereas XLM-RoBERTa employs a SentencePiece-based subword segmentation scheme, which has been shown to provide more flexible and robust handling of rich morphology and orthographic variation. For a morphologically complex and low-resource language such as Armenian, this difference in tokenization is expected to play a crucial role in reducing subword fragmentation and improving the consistency of lexical representations.

Both BERT-Multilingual Uncased and XLM-RoBERTa clearly outperform the two Armenian-specific baselines, achieving identical F1 scores of 52%. Their advantage appears to stem from their larger parameter counts and robust multilingual pretraining, which allow them to capture contextual and morphological variation more effectively than lighter models. In contrast, HyeBert and spaCy exhibit considerably lower precision and recall, reflecting their limited ability to generalize across the sparse and morphologically complex terminology characteristic of Armenian epigraphy. These results suggest that models with broader multilingual exposure and deeper contextual representations are better suited to this task than smaller or more rigid architectures.

3.5.5 Complete Dataset Implementation

Attempting this experiment on a restricted dataset highlighted the need for a fairly generalizable model that can adapt to scarce, context-dependent data points. Having established a first indicative baseline on DHV-10, the next step would be to train and evaluate the previously tested model on a more complete dataset.

This novel dataset, is formed by labelling sen-

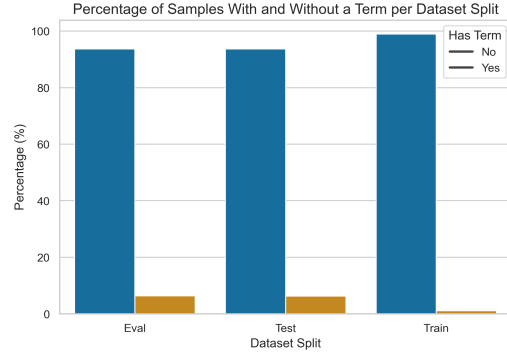


Figure 11: Distribution of sentences across dataset splits for the complete dataset in percentages.

tences extracted from 20 different books (highlighted in grey in Table 1), drastically changing the experiment’s scale: the baseline was established on around 1200 sentence, compared to more than 56,000 sentences at this stage, not only broadening the range of seem terms but also adding a significant number of sentence that do not contain epigraphy-related terms as depicted by Figure 11. It is important to note that neither the evaluation nor the testing datasets were changed to maintain consistency in results subject to different experimental settings.

The results presented in Table 6 suggest that the augmentation to a fuller dataset pushed the models to detections that were both more complete and more accurate, as evidenced by the improvement in evaluation metrics. Furthermore, these results reinforce the idea that multilingual transformer models are better suited for this task, since mBERT and XLM-Roberta clearly outperformed the other baselines. Finally, the most striking aspect of these findings is the marked improvement in mBERT’s ability to detect inscription-related terms, surpassing XLM-Roberta. This can be explained by the specific characteristics of the dataset: with only around one percent of the total sentences containing terms of interest, the task is highly imbalanced and relies heavily on the model’s capacity to generalize from sparse positive examples.

Model	# Params	Languages	Layers	Hidden Size
mBERT (BERT-Multilingual)	~110M	102	12	768
XLM-RoBERTa (Base)	~270M	100	12	768

Table 5: Technical characteristics of the transformer models retained for evaluation (Conneau et al. [2019], Devlin et al. [2018]).

Model Name	Precision	Recall	F1
Bert Multilingual Uncased	0.82	0.50	0.63
XLM-Roberta	0.67	0.44	0.53
Spacy	0.50	0.06	0.11
HyeBert	0.50	0.06	0.10

Table 6: Performance of all models on the complete dataset.

mBERT’s subword vocabulary and pretraining appear better aligned with the morphology and tokenization patterns encountered in Armenian epigraphic terms, allowing it to recover more relevant entities under low-signal conditions.

4 Results

The model developed is ultimately tested on a set of completely unseen books to determine its capacity to detect familiar terms and scale to new terms. The new data is passed through the cleaning pipeline and split into sentences. Using the model for inference shows that it was able to adapt to the grammatical flexibility of the Armenian language by extracting the in-context versions of the previously seen data. However, when it came down to completely novel terms, the model was unable to discover out-of-the-box terms.

Figure 12 provides a quantitative comparison between the reference, benchmark, and inferred term sets. The benchmark set contains 79 unique terms, while the inferred set contains 53, of which 40 terms (75.5% of the inferred set) overlap with the benchmark. This indicates that the model is highly effective at recovering benchmark term types when applied to unseen corpora.

In contrast, the reference set contains 37 unique terms, of which only 12 overlap with the benchmark and none intersect with the inferred set.

The absence of any direct intersection between the reference and inferred terms quantitatively confirms that the model pains to discover genuinely novel terms outside of the supervised benchmark space. Instead, its strength lies in scaling known term types across new grammatical contexts and previously unseen documents,

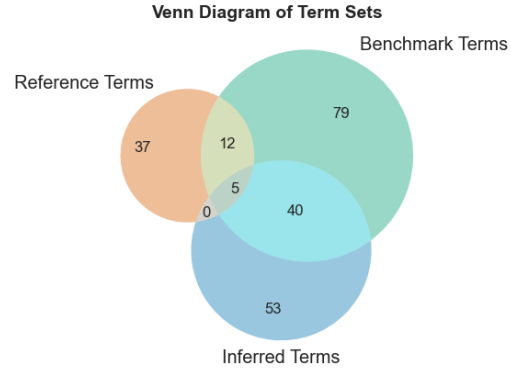


Figure 12: Venn Diagram of Reference, Benchmark and Inferred Terms.

demonstrating strong contextual generalization but limited out-of-vocabulary discovery.

5 Conclusion and Future Work

This study primarily serves as a baseline for systematic inscription-type terminology extraction using modern solutions. While transformer-based NER serves for detecting grammatical adaption of previously-seen terminology, it lacks when it comes to extrapolating context to find completely unseen terminology. A fully-automated, hands-free approach relying solely on NER is hence not envisaged at this time, further highlighting the role of experts in human-backed pipelines. On the other hand, it is also important to mention the work that was set in place before the NER step, as it is a robust foundation for developing more complex solutions to the problem at hand: the systematic cleaning, labeling and detection set the ground work for contextual validation, as they transform fairly different data into a homogeneous, machine-interpretable output.

5.1 Future Work

Surface-level extraction proved insufficient for the complexity that the Armenian language presents, putting forth the need to shift towards a context-heavy approach, which can be tackled using different context-modeling techniques: from distributional semantics and collocation

analysis, to sentence-level disambiguation, hierarchical concept induction, and cross-ontology alignment, these methods could shift the analysis from a pragmatic word-hunt to a more causal, context-backed mode of detection. Finally, aligning the Armenian lexicon with EA-GLE and FAIR Epigraphy ontologies provides a formal mechanism for semantic interoperability, allowing Armenian inscription data to be integrated into multilingual knowledge graphs, compared across corpora, and queried using standardized conceptual categories.

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A Prompting

Listing 1: Zero-shot prompt used for Armenian term extraction

```
1 zero_shot_prompt = f"""
2 You are an expert in Armenian epigraphy.
3 Your task is to extract specific terms from the provided text chunks.
4 For each term, you will identify its occurrences in the text, along
5 with the context in which it appears.
6 The context should include a few words before and
7 after the term to provide clarity on its usage.
8
9 Each occurrence should be documented with the following details:
10 - file_name: The name of the file where the term was found.
11 - page_number: The page number in the document where the term was found.
12 - term: The term itself.
13 - context: A snippet of text surrounding the term to provide context.
14
15 Each term can have multiple occurrences, and each occurrence should be recorded separately and stored in
16 an Occurrence object.
17 All occurrences of a term should be grouped under a Term object, which includes the term, its definition,
18 and a list of its occurrences.
19
20 Finally, all Term objects should be compiled into an ExtractionResult object.
21
22 The text is written in Armenian and may require special handling.
23 Extract all occurrences of the following terms from the document text.
24 Terms and definitions: {terms}
25
26 --- BEGIN TEXT CHUNK ---
27 {chunk}
28 --- END TEXT CHUNK ---
29 """
```